RPD-Based Collision-Free Path Planning for Autonomous Vehicles

Jingyun Xu, Shichun Yang, *Senior Member, IEEE*, Bin Sun, Tianyang Gong, Xinjie Feng, and Yaoguang Cao

***Abstract*—When autonomous vehicles (AVs) encounter sudden hazards, how to take the correct collision-free measures in emergencies has always been a focus of research. The reachable sets method has been widely applied in local path planning methods for AVs to address the collision-free path planning problem for dense, dynamic obstacles in traffic scenarios. However, the reachable sets method is limited by its inherent tendency to overestimate obstacles, making it difficult to quantify the probability of conflicts between other traffic participants and AVs. Furthermore, path planning within the reachable set may encounter local minima, potentially resulting in planning failure.This paper proposes a reachable probabilistic distribution (RPD) method, integrating reachable sets with the Interactive Multiple Model Filtering (IMM) method. The proposed method begins by collecting surrounding environmental data using Vehicle-to-Everything (V2X) technology. It then models both the obstacles and the ego vehicle, followed by constructing the occupancy sets of the obstacles and the reachable set of the ego vehicle. IMM then predicts obstacle behavior to determine their probabilities of being in each lane at future intervals. These probabilities are combined with obstacles' occupancy sets and ego vehicle's reachable sets to derive RPD. Compared to traditional reachable sets, the RPD framework leverages probabilistic information to simplify candidate path evaluation, greatly enhancing computational efficiency. To avoid local minima failures in local path planning within RPD, a method combining discrete strategies with Model Predictive Control (MPC) is proposed. The proposed method offers a safety reference for intelligent transportation systems based on V2X technology. It employs discrete strategies to achieve comprehensive coverage of feasible local paths, while MPC is used for real-time tracking. Validated in AVs, this method ensures robust collision avoidance with stability and safety, demonstrating its applicability in intelligent transportation systems.**

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I. Introduction

IN RECENT years, autonomous vehicles (AVs) have gained widespread attention for their potential to transform transportation systems. However, real-world roads are complex and unpredictable, requiring AVs to ensure safety under imminent hazard conditions. International standards such as ISO 21448 and R157 have been established, detailing AVs' Safety of the Intended Functionality (SOTIF) and setting higher safety requirements. Therefore, this study focuses on formulating decision-making and path planning strategies for AVs in high-risk scenarios. The foundation of this research is built on V2X technology, which provides real-time perception of the surrounding traffic environment to allow accurate path planning and ensure safety control.

Traditional path planning methods, such as Rapidly-exploring Random Trees (RRT) and Lattice, are often used to create collision-free paths but may need help in unpredictable environments with sensor inaccuracies [1], [2], [3]. Therefore, robust and adaptive methods are required. For instance, Chai et al. proposed a multi-layer approach for trajectory planning in overtaking [4], while Ji et al. introduced an artificial potential field approach that effectively addressing the local minimum problem [5]. Additionally, machine learning-based methods have gained significant attention. Chai et al. decomposed the planning task into layers handled by neural networks and achieved favorable results [6].

In addition to the abovementioned methods, approaches based on reachable sets also show significant potential in collision-free trajectory planning tasks. These approaches consider all possible states a dynamic system can reach from a given initial state, making them particularly well-suited for path planning in complex traffic scenarios. Reachable set-based methods have been widely used in robotic path planning tasks [7], [8], [9], and their application to vehicles has garnered considerable interest from researchers. However, the computation of reachable sets is highly varied and can consume substantial resources when detailed models are used, requiring significant computing power. Although some studies have optimized the computational efficiency of reachable sets [10], the impact of computational demands remains a key consideration.

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Over-conservatism in emergency collision-free path planning can result from simplified linear models of obstacles, reducing available planning space and resulting in overly cautious maneuvers. Gharavi et al. introduced a nonlinear MMPS approach to reduce conservatism in Stochastic Model Predictive Control (MPC), highlighting the need for discrete modeling to capture the behavior of moving obstacles better and address over-conservatism [11]. Additionally, the impact of local minima on optimization tasks for path planning also warrants attention, as AVs may stop at unintended positions. Introducing virtual target points can reduce this risk [12], but further research is needed to prevent local minima in reachable set-based optimization.

This study applies the Interactive Multiple Model (IMM) method within reachable sets to address the over-conservatism problem by constructing a Reachable Probabilistic Distribution (RPD) model. IMM could predict obstacle strategies, assigning high probabilities to likely paths and providing AVs with richer decision-making information. A MPC method employing a discrete strategy is also proposed to address the issue of local minima by providing multiple collision-free reference lines. Compared to traditional reachable set methods, using discrete strategies for reference line planning within the RPD is more efficient, as the computational process for evaluating various lines is significantly simplified. These methods are combined to enhance emergency planning and have been validated in simulations and on-road tests for highway scenarios. The contributions of this study are three-fold:

- We developed a new model, combining reachable sets and IMM to derive the RPD of obstacles.
- We designed a path planning and vehicle control method that combines discrete strategies with the MPC.
- We deployed this method on an AV to verify the feasibility of this method and processed experimental data.

It should be noted that the current study focuses on high-speed highway scenarios with relatively large curvature radii. Urban driving scenarios involving low-speed high-curvature turns are not fully addressed, as the vehicle kinematic model used in this work does not account for tire dynamics and lateral slip effects under extreme steering inputs.

This paper will present some related works in Chapter II. Chapter III will introduce the method of this work, where the construction methods of obstacles' occupancy sets and AV's reachable sets will be explored in Section III-A, the construction method of RPD using the IMM will be presented in Section III-B, and the planning method for the collision-free reference line will be presented in Section III-C, and the vehicle control method based on the MPC will be presented in Section III-D. The validation of this study will be presented in Section IV, with IV-A showcasing the simulation results and IV-B presenting the vehicle experiment results. Finally, the conclusions and future prospects of this paper will be presented in Chapter V.

II. Related Works

The application of reachable sets in AV path planning has been a popular research topic, with many studies contributing to this area. Matthias and his team have made significant contributions. In their earlier work, the initial set was determined by a nonlinear system with uncertain parameters and inputs, with the error calculated by linearizing the Lagrange cosine term [13]. Zonotopes were used to represent reachable sets in previous studies, but their limitations became apparent when dealing with transition preservation, which defines the discrete behavior of hybrid systems. To address this, the team switched to using polytopes to represent reachable sets [14]. Later, over-conservatism was introduced to compute reachable sets under uncertain initial states and input conditions [15].

The method was validated on real data in [16] and [17], and the team developed SPOT for predicting traffic participants [18]. In [19], they proposed a simplified method for constructing AV drivable regions by combining forward and backward solutions to compute reachable sets. Additionally, they introduced a method to compute reachable sets of road vehicles considering dynamic obstacles, reducing over-conservatism by using interconnected reachable sets propagating from multiple time points. In [20], reachable sets and convex optimization methods were combined for path planning in traffic scenarios. The safety assurance method proposed in [21] makes the reachable set-based path planning approach more reliable after path planning failure.

Other research directions on reachable sets in path planning have also emerged. For instance, [22] made the reachability analysis of large-scale systems feasible, reducing the performance gap in the reachability analysis of linear systems. In [23] and [24], reinforcement learning was incorporated into reachable sets, and its feasibility was verified. Furthermore, the method was optimized by considering occlusion and traffic rules [25], improving its applicability. The team also developed AROC, an automated optimal controller-integrated toolkit based on reachable sets [26], facilitating research in autonomous driving.

Other teams have also explored reachable sets for path planning. For example, Patsko et al. in [27] studied the feasibility of reachable sets in observing the motion of the Dubins car. W et al. proposed a cost-constrained model of reachable sets using a value function and defined cost-constrained reachable sets in [28]. Zheng developed a flexible and efficient fully symmetric tube-based robust model predictive controller based on reachable sets [29]. These studies show the high value of reachable sets obtained through reachability analysis for path planning in AV road traffic scenarios.

MPC in intelligent transportation systems, especially for autonomous vehicle (AV) decision-making and motion planning, requires robust strategies to manage uncertainties. Recent research reflects this focus. Gu et al. developed a tube-based event-triggered stochastic MPC to handle additive uncertainties in [30], balancing computational efficiency with control performance. Similarly, Yin et al. employed a distributed stochastic MPC framework with Taguchi's robustness to improve vehicle platoon stability under disturbances in [31]. These advancements demonstrate the ongoing development of MPC frameworks to enhance the reliability and safety of AVs in complex traffic environments.

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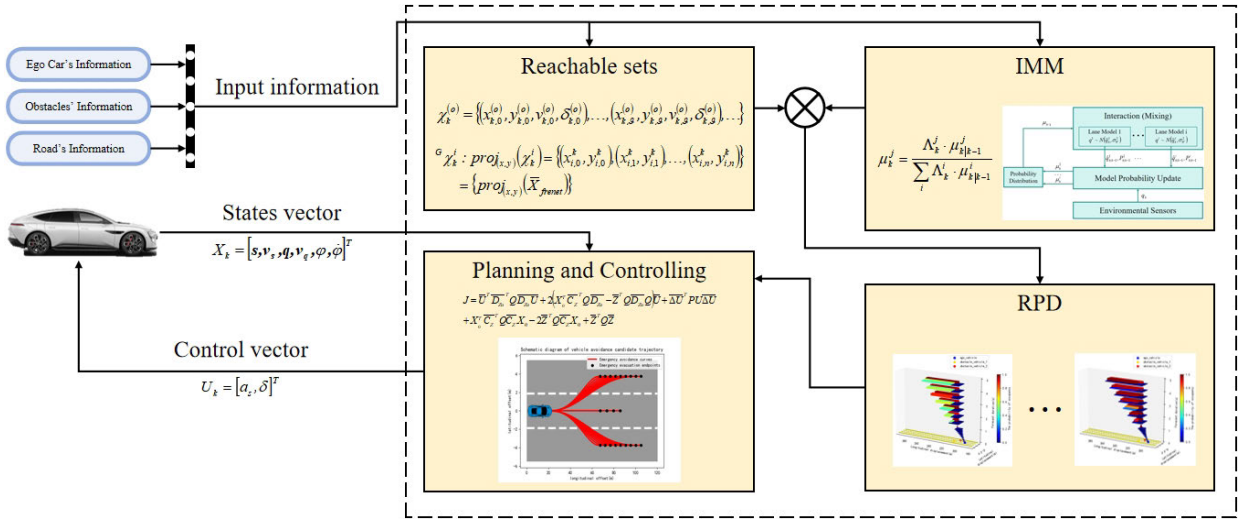

Fig. 1. The collision-free path planning framework based on RPD and MPC.

III. Method

Inspired by desensitized trajectory optimization in [6], this study integrates IMM and reachable set models to construct the RPD for AVs, combining it with variable acceleration strategies and MPC for path planning and control. The method involves four steps. First, calculate occupancy sets for all traffic participants. Second, IMM will be used to predict lane occupancy probabilities, forming the RPD with the AV's reachable set. Third, generate and select the optimal reference trajectory based on variable acceleration strategies and the RPD. Finally, MPC is applied to follow the optimal trajectory and avoid obstacles. The framework is shown in Fig. 1.

A. Constructing Reachable Sets and Occupancy Sets

Constraints must be imposed on the drivable road space before creating reachable sets and occupancy sets. The drivable road space $\mathcal{R}$ for the ego vehicle is defined as a nonlinear function integrating sensor-derived environmental perception with measurement uncertainties, expressed as Eq. (1):

$$\mathcal{R} = f(\mathbf{x}_{\text{ego}}, \mathcal{S}, \mathcal{T}), \tag{1}$$

where $\mathbf{x}_{\text{ego}} = (x, y, v, \psi)$ is the ego vehicle state in Frenet coordinates, $\mathcal{S}$ denotes sensor-based environmental perception including obstacle positions with uncertainties, and $\mathcal{T}$ represents traffic rule constraints. Specifically, $\mathcal{R}$ is constructed as Eq. (2):

$$\mathcal{R} = \Big\{(s, q) \in \mathbb{R}^2 \mid q \in [-\hat{w}(s) - \epsilon_w(s), \hat{w}(s) + \epsilon_w(s)] \\ \wedge (s, q) \notin \mathcal{F} \wedge \mathbf{obs}_i + \mathcal{E}_i \subseteq \mathcal{X}_{\text{free}}\Big\}, \tag{2}$$

here, s represents the longitudinal distance from a point to the projection onto the current road's central axis, and q denotes the normal distance from the point to the road's central axis, $\hat{w}(s)$ is the measured lane width at position s, $\epsilon_w(s)$ quantifies the sensor uncertainty in road boundary detection, $\mathcal{F}$ is the set of forbidden regions, $\mathbf{obs}_i$ is the detected obstacle position, and $\mathcal{E}_i$ is the sensor error ellipsoid. The nonlinearity of $f(\cdot)$ captures the complex mapping from noisy sensor inputs to drivable regions:

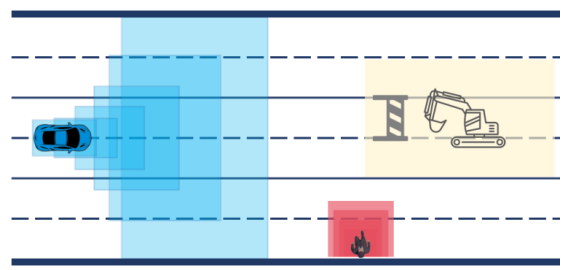

Fig. 2. The occupancy sets for different types of obstacles.

$$\mathcal{R} = \left\{(s, q) \mid (s, q) \in \mathcal{X}_{\text{road}} \wedge \frac{\partial f}{\partial \mathcal{S}} \neq 0\right\}. \tag{3}$$

Then, construct reachable sets and occupancy sets. Dynamic obstacles, unlike static ones fixed in a rectangular space, require initial modeling. Pedestrians are represented by mass models, while vehicles and bicycles use bicycle dynamics. Initial occupancy sets are defined as vectors, explicitly incorporating sensor measurement uncertainties to encompass all possible states of obstacles under sensing errors. These sets are propagated over time with bicycle models to generate possible obstacle state vectors for each time interval and are projected onto a 2D plane to determine occupancy sets. Fig. 2 demonstrates the varied methods for constructing these sets across different obstacle types.

In terms of the specific constructing of the kinematic model, the bicycle model is described as depicted in Fig. 3. Supposing dynamic obstacles (typically obstructive vehicles) adopt the bicycle model, with their state vector denoted as $(x\ y\ w\ \psi)$, the bicycle models for this category of obstacles are described as follows:

$$x^{(o)}_{\vartheta,k+1} = x^{(o)}_{\vartheta,k} + v^{(o)}_{\vartheta,k} \cdot \cos(\psi^{(o)}_{\vartheta,k} + \beta) \cdot k\Delta t, \tag{4a}$$
$$y^{(o)}_{\vartheta,k+1} = y^{(o)}_{\vartheta,k} + v^{(o)}_{\vartheta,k} \cdot \sin(\psi^{(o)}_{\vartheta,k} + \beta) \cdot k\Delta t, \tag{4b}$$

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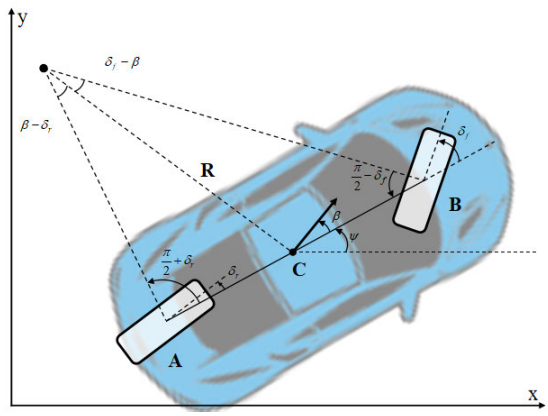

Fig. 3. The vehicle kinematic mode.

$$\psi^{(o)}_{\vartheta,k+1} = \psi^{(o)}_{\vartheta,k} + v^{(o)}_{\vartheta,k} \cdot \sin(\beta_\vartheta) \cdot k\Delta t / l_r, \tag{4c}$$

$$v^{(o)}_{\vartheta,k+1} = v^{(o)}_{\vartheta,k} + a_\vartheta \cdot k\Delta t. \tag{4d}$$

where x represents the longitudinal position of the vehicle, y denotes the latitudinal position, ψ signifies the vehicle's yaw angle, v indicates the velocity, Δt stands for the sampling time interval, and β_ϑ represents the front wheel steering angle of the vehicle. The specific description of the front wheel steering angle of the vehicle is: $\beta_\vartheta = \tan^{-1}\left(\frac{l_r}{l_f+l_r}\tan(\delta_{\vartheta,f})\right)$, where ϑ represents the strategy.

Then, the abovementioned occupancy set constructed for obstacle vehicles could be represented as: $O^{(i)}_k = \{\xi^{(i)}_k \in \Xi^{(i)}_k\}$, where $\xi^{(i)}_k$ represents a possible state of the vehicle at the k-th time interval, and $\Xi^{(i)}_k$ denotes each state the vehicle can reach at the k-th time interval. In particular, future trajectories of dynamic obstacles are seamlessly integrated into their occupancy sets. By continuously monitoring these trajectories in real-time, the method effectively eliminates response lag caused by sudden path changes, thereby enabling autonomous vehicles to make timely and safe decisions even in complex traffic environments.

B. RPD: Probability Distributions Integrated With IMM and Reachable Sets

Although the occupancy set considers all possible states of obstacles during construction, it assumes equal probability for all strategies the obstacles may take. This assumption does not align with real-world conditions, resulting in over-conservation occupancy sets. Therefore, incorporating IMM into the reachable set might be helpful.

Specifically, IMM could predict the probability of obstacle occurrence on each road segment at each future time step. By modifying the original occupancy set this way, the updated set, when projected onto the Frenet coordinate system, exhibits a probability distribution feature that aligns with road-based predictions. The specific process is as follows.

1) Project the Occupancy Sets Onto the Frenet Coordinate System: For computational convenience, it is necessary to project the occupancy sets onto the Frenet coordinate system for analysis. The state vector of the vehicle in Cartesian coordinates, $\overline{X}_{cartesian} = (x, y, \theta_x, k_x, v, a)^T$, is transformed into the state vector in the Frenet coordinate system, $\overline{X}_{frenet} = (s, \dot{s}, \ddot{s}, q, q', q'')^T$, taking the center line as the base frame curve.

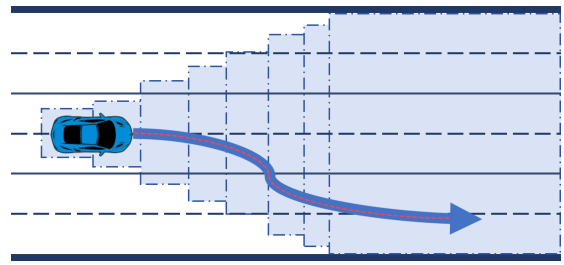

Fig. 4. Diagram of vehicle reachable sets.

In Cartesian coordinates, the components of the state vector represent the vehicle's x-axis coordinate, y-axis coordinate, heading direction, current curvature, current velocity, and current acceleration. In contrast, the components of the state vector in the Frenet coordinates represent the vehicle's longitudinal coordinate, velocity along the base frame, acceleration along the base frame, Frenet latitudinal coordinate, derivative of the Frenet latitudinal coordinate concerning the longitudinal coordinate, and second derivative of the Frenet latitudinal coordinate concerning the longitudinal coordinate. To achieve the transformation of obstacle states from the Cartesian coordinate system to the Frenet coordinate system, a projection operator is introduced as $proj_\iota(x)$. This operator signifies the extraction of element ι from vector x, as exemplified in $proj_{(m_1,m_3)}((m_1, m_2, m_3)^T) = (m_1, m_3)^T$.

When projecting the occupancy sets, the boundary constraints of the drivable road space $\mathcal{R}$ must be satisfied. Then, obstacle occupancy sets $\chi^{(o)}_k$ are initialized in Eq. (5):

$$^G\chi^{(o)}_{k,0} = \left\{\xi^{(i)}_k \in \Xi^{(i)}_k \mid \mathrm{proj}_{(s,q)}(\xi^{(i)}_k) \in \mathcal{R} \oplus \mathcal{E}_{\mathrm{sen}}\right\}. \tag{5}$$

Eq. (6) represents the expression of occupancy sets of obstacle vehicles in the Cartesian coordinate system.

$$\chi^{(o)}_k = \left\{\left(x^{(o)}_{k,0}, y^{(o)}_{k,0}, v^{(o)}_{k,0}, \delta^{(o)}_{k,0}\right), \left(x^{(o)}_{k,1}, y^{(o)}_{k,1}, v^{(o)}_{k,1}, \delta^{(o)}_{k,1}\right), \ldots, \left(x^{(o)}_{k,\vartheta}, y^{(o)}_{k,\vartheta}, v^{(o)}_{k,\vartheta}, \delta^{(o)}_{k,\vartheta}\right), \ldots\right\}, \tag{6}$$

where ϑ represents the number of strategies, o represents the number of vehicles, and k represents the time interval. The occupancy sets of the vehicle o theoretically contain an infinite number of state vectors at the k-th time interval. Next, considering that the occupancy set is a collection of multiple states, for the sake of computational convenience, the set of reachable obstacle vehicles in the Frenet coordinate system can be represented as multiple sets depending on the results in the Cartesian coordinate system.

The process of projecting the occupancy set of obstacle vehicles onto a two-dimensional Frenet plane can be described by:

$$^G\chi^i_k = proj_{(s,q)}(\chi^i_k) = \left\{\left(s^i_{k,0}, q^i_{k,0}\right), \left(s^i_{k,1}, q^i_{k,1}\right), \ldots, \left(s^i_{k,n}, q^i_{k,n}\right)\right\} = proj_{(s,q)}\left(\overline{X}_{frenet}\right). \tag{7}$$

The reachable sets projected onto the Frenet plane can be obtained at this point, as shown in Fig. 4, these sets

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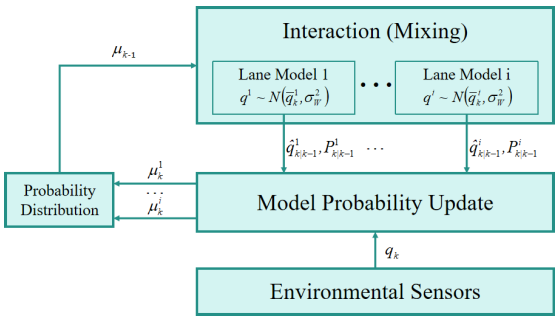

Fig. 5. Schematic diagram of the IMM framework used in this method.

overestimates the operating states of all obstacle vehicles. Consequently, the occupancy set at each moment fully covers the positions of the obstacle vehicles, regardless of their exact locations or driving strategies. For instance, if a vehicle is currently executing a right lane-change strategy, the entire set of trajectory points for this maneuver is included within the occupancy sets. However, many occupancy sets remain unused, representing an evident inefficiency and resulting in an over-conservative collision avoidance strategy.

2) Construct the IMM Filter: Motivated by [32], IMM can be used to predict the lane of each obstacle vehicle at each future time interval. By characterizing the uncertain motion of vehicles in the Frenet coordinate system, it is possible to estimate the probability of a vehicle occupying different lanes in the future. The likelihood of a vehicle converging to a particular lane can be calculated based on its lateral position q in the Frenet coordinates. In [32], the authors proposed a method for dynamically adjusting conditional transition probabilities. However, our experimental results indicate that, while this method offers minimal advantages over using a constant transition probability, it significantly increases computational time. Therefore, we opted to use continuous transition probabilities. Fig. 5 illustrates a schematic of the IMM framework used in this context.

Specifically, the fusion correlation coefficients are computed based on a predefined correlation matrix. This matrix, established beforehand, serves to quantify the degree of correlation between each model. The methodology for calculating the fusion correlation coefficient $\lambda^{j}_{k|k-1}$ is shown as:

$$\lambda^{ij}_{k|k-1} = \frac{p_{ij} \cdot \lambda^{i}_{k-1}}{\sum_{i} p_{ij} \cdot \lambda^{i}_{k-1}}, \tag{8}$$

where p_{ij} denotes the correlation coefficient between the i-th and j-th models, which, in the context of this work, represents the correlation coefficient between the i-th and j-th lanes counted from left to right. $\lambda^{ij}_{k|k-1}$ is computed to signify the mixed-model prediction probability of vehicles transitioning from lane i to lane j.

Next, the mixed latitudinal offset of vehicles is computed by taking the weighted sum of each lane, primarily achieved by multiplying the mixed probability model obtained from the previous step by the latitudinal offset of the center line, as shown in:

$$\hat{q} = \sum_{i} \lambda^{ij}_{k|k-1} \cdot \bar{q}^{i}_{k-1}, \tag{9}$$

$$P^{j}_{k|k-1} = \sum_{i} \lambda^{ij}_{k|k-1} \cdot \lfloor Q^{j} + (\bar{q}^{i}_{k-1} - \hat{q}^{i}_{k|k-1}) \times (\bar{q}^{i}_{k-1} - \hat{q}^{i}_{k|k-1})^{T} \rfloor, \tag{10}$$

where $Q^{j} \sim N\left(0, \sigma^{2}_{w}\right), j \in [1, 2, \ldots, N_l]$, the variance of this normal distribution is $\sigma^{2}_{w} = \frac{1}{16} L^{2}_{lane_width}$, where N_l is the number of lanes and L_{lane_width} is the width of lane.

The probability model is updated using the residual r_k and the covariance S_k at the time k, and the residual term r_k is computed as the difference between the observed latitudinal position deviation of the vehicle at the time step k and the mixed latitudinal deviation, as shown in:

$$r^{j}_{k} = z_k - \hat{q}^{j}_{k|k-1}, \tag{11}$$

where $z_k = q_k + blue\omega$.

$$S^{j}_{k} = P^{j}_{k|k-1} + \omega, \tag{12}$$

where ω is the zero-mean white Gaussian measurement noise of the latitudinal offset. Additionally, assuming the likelihood function of the observations follows a Gaussian probability density function (PDF) in Eq. (14), the model probabilities for each lane are updated accordingly.

$$\Lambda^{j}_{k} = N\left(r^{j}_{k}; 0, S^{j}_{k}\right) = \frac{1}{\sqrt{2\pi \cdot S^{j}_{k}}} \exp\left[-\frac{\left(r^{j}_{k}\right)^{2}}{2S^{j}_{k}}\right], \tag{13}$$

$$\mu^{j}_{k} = \frac{\Lambda^{j}_{k} \cdot \mu^{j}_{k|k-1}}{\sum_{i} \Lambda^{i}_{k} \cdot \mu^{i}_{k|k-1}}. \tag{14}$$

Consequently, the probability distributions of the vehicle's lane at each sampling time interval can be computed. It can be seen that the time complexity of IMM is $O\left(n^{2}_{state}\right)$, which is related to the number of states.

3) RPD Construction: Combine the IMM with the reachable sets to construct the RPD. First, as discussed above, the IMM provides probabilities for each behavior mode adopted by obstacles, with these modes closely tied to specific lanes. Consequently, the probability of obstacles occupying each lane over a future time horizon could be derived. We then divide the occupancy sets of obstacles into several axial rectangles based on lane boundaries and assign the model-predicted probabilities of the obstacle being in each lane at each time step to these axial rectangles:

$$^{G}\chi^{i}_{k} = \left\{\left(s^{i}_{k,0}, q^{i}_{k,0}, \overline{\mu}^{i}_{k,0}\right), \left(s^{i}_{k,1}, q^{i}_{k,1}, \overline{\mu}^{i}_{k,1}\right), \ldots, \left(s^{i}_{k,\vartheta}, q^{i}_{k,\vartheta}, \overline{\mu}^{i}_{k,\vartheta}\right), \ldots\right\}. \tag{15}$$

Fig. 6 illustrates a schematic diagram of the probability distribution of the obstacle-occupancy sets. Different colors represent the magnitude of the probability of occupancy in each region. The areas filled in red indicate that obstacles are most likely to appear at that moment, while green indicates the lowest probability of occupancy. Yellow represents the probability of occupancy lying between the two areas mentioned above. After probabilistic representation, the obstacles'

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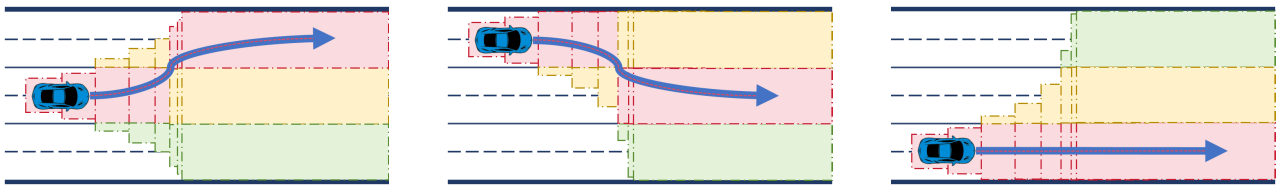

(a) Vehicle lane change to the left (b) Vehicle cut in the middle lane (c) Vehicle proceeding straight in the right lane

Fig. 6. Integration of IMM and reachable sets for establishing RPD.

occupancy sets will possess more excellent predictive value, providing a more rational and efficient basis for the path planning section.

Second, we calculate the drivable sets for the ego vehicle in Frenet coordinates, denoted as:

$$ {}^{G}D_k^{ego} : proj_{(s,q)}(D_k^{ego}) = \left\{\left(s_{k,0}^{ego}, q_{k,0}^{ego}\right), \left(s_{k,1}^{ego}, q_{k,1}^{ego}\right), \ldots, \left(s_{k,n}^{ego}, q_{k,n}^{ego}\right)\right\}. \tag{16} $$

These drivable sets are derived by considering only road boundary constraints and the kinematics of the ego vehicle. All subsequent strategies will be encompassed within these drivable sets.

Third, based on the intersection of ${}^{G}D_k^{\text{ego}}$ and ${}^{G}\chi_k^{(o)}$, identify the overlap between ${}^{G}D_k^{\text{ego}}$ and the occupancy sets ${}^{G}\chi_k^{(o)}$ of all obstacles. For each point in the intersection, add the model prediction probability of that point in ${}^{G}\chi_k^{(o)}$ to ${}^{G}D_k^{\text{ego}}$, resulting in the reachable probability distribution ${}^{G}R_k^{\text{ego}}$ for the ego vehicle. ${}^{G}R_k^{\text{ego}}$ is shown as:

$$ {}^{G}R_k^{ego} = \left\{\left(s_{k,0}^{ego}, q_{k,0}^{ego}, \overline{\mu}_{k,0}^{ego}\right), \left(s_{k,1}^{ego}, q_{k,1}^{ego}, \overline{\mu}_{k,1}^{ego}\right), \ldots, \left(s_{k,\vartheta}^{ego}, q_{k,\vartheta}^{ego}, \overline{\mu}_{k,\vartheta}^{ego}\right), \ldots\right\}. \tag{17} $$

The parameter $\overline{\mu}_{k,\vartheta}^{ego}$ is influenced by the first two components of its state vector, assuming that they represent the position in lane i. Whether this position conflicts with other obstacle vehicles' occupancy sets must be observed separately. The obstacle vehicle's number is stored in $obstacle_set_{k,\vartheta}$ if a conflict occurs. This process is described as:

$$ \overline{\mu}_{k,\vartheta}^{ego} = \frac{\sum^{(o)} \mu_k^i}{n_{obstacle}}, (o) \in obstacle_set_{k,\vartheta}, \tag{18} $$

where $n_{obstacle}$ is the number of the obstacle vehicles.

Finally, collision-free decisions and path planning can be made in the RPD ${}^{G}R_k^{\text{ego}}$. ${}^{G}R_k^{\text{ego}}$ represents a collection of multiple states along with their conflict probabilities with obstacles, which is not highly practical for detailed path planning. Instead, dividing ${}^{G}R_k^{\text{ego}}$ into several axis-aligned rectangles along the boundaries of the occupied regions of obstacles and lane lines are considered more practical. For each time interval in ${}^{G}R_k^{\text{ego}}$, there are several axis-aligned rectangles, denoted as $Rect_k^{\zeta}$, where ζ represents the rectangle index. The risk value $Risk_k^{\zeta}$ of each rectangle is proportional to the conflict probability $\overline{\mu}_{k,\vartheta}^{\text{ego}}$ at each point ϑ within it. Normalizing the risk values of all rectangles at each time step is important.

C. The Collision-Free Reference Line Planning via Discrete Strategies

To efficiently provide critical decision-making information for the subsequent MPC control algorithm while minimizing computational time, this work introduces a collision-free reference path generation method inspired by the two-stage trajectory planning and control approach in [6]. This method leverages a set of predefined discrete strategies, where each element corresponds to an adaptive maneuver. Bézier curves, renowned for their excellent flexibility and smoothness properties, are utilized to generate obstacle-avoidance paths based on the ego vehicle's current state and lane information. A Bézier curve, defined by $n+1$ control points $\mathcal{P}_0, \mathcal{P}_1, \mathcal{P}_2, \cdots, \mathcal{P}_n$ can be expressed as:

$$ B(t) = \sum_{i=0}^{n} \frac{n!}{i!(n-i)!} t^i (1-t)^{n-i} \mathcal{P}_i, t \in [0,1]. \tag{19} $$

By modifying the original formula of the Bézier curve, the functional expression for each alternative collision-free reference line can be obtained as:

$$ B(t) = \sum_{i=0}^{5} \frac{5!}{i!(5-i)!} \left(\frac{t}{T}\right)^i \left(1-\frac{t}{T}\right)^{5-i} \mathcal{P}_i, t \in [0,T], \quad i = 0,1,2,3,4,5. \tag{20} $$

This method utilizes a fifth-order Bézier curve. The distribution functions of speed, acceleration, and jerk for the ego vehicle are derived as follows:

$$ \frac{d}{dt}B(t) = \sum_{i=0}^{4} \frac{4!}{i!\,(4-i)!} \left(\frac{t}{T}\right)^i \left(1-\frac{t}{T}\right)^{4-i} \frac{5}{T} \cdot \left(\mathcal{P}_{i+1} - \mathcal{P}_i\right), t \in [0,T], \tag{21} $$

$$ \frac{d^2}{dt^2}B\left(t\right) = \sum_{i=0}^{3} \frac{3!}{i!\,(3-i)!} \left(\frac{t}{T}\right)^i \left(1-\frac{t}{T}\right)^{3-i} \frac{20}{T^2} \cdot \left(\mathcal{P}_{i+2} - 2\mathcal{P}_{i+1} + \mathcal{P}_i\right), t \in [0,T], \tag{22} $$

$$ \frac{d^3}{dt^3}B(t) = \sum_{i=0}^{2} \frac{2!}{i!(2-i)!} \left(\frac{t}{T}\right)^i \left(1-\frac{t}{T}\right)^{2-i} \frac{60}{T^3} \cdot \left(\mathcal{P}_{i+3} - 3\mathcal{P}_{i+2} + 3\mathcal{P}_{i+1} - \mathcal{P}_i\right), t \in \left[0,T\right]. \tag{23} $$

The relationship between the acceleration derivative distribution control points and each control point of the

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collision-free reference line is given by:

$$\begin{cases} \mathcal{P}_0''' = \dfrac{60}{T^3}\left(\mathcal{P}_3 - 3\mathcal{P}_2 + 3\mathcal{P}_1 - \mathcal{P}_0\right) \\ \mathcal{P}_1''' = \dfrac{60}{T^3}\left(\mathcal{P}_4 - 3\mathcal{P}_3 + 3\mathcal{P}_2 - \mathcal{P}_1\right) \\ \mathcal{P}_2''' = \dfrac{60}{T^3}\left(\mathcal{P}_5 - 3\mathcal{P}_4 + 3\mathcal{P}_3 - \mathcal{P}_2\right). \end{cases} \tag{24}$$

Assuming the initial heading angle of the vehicle is zero and at the end of the control process, the vehicle's lateral velocity and acceleration are both zero, the longitudinal acceleration should be uniform. The control points for the acceleration derivative distribution function are:

$$\begin{cases} \mathcal{P}_0''' = \left(0, \dfrac{60L_{offset}}{T^3}\right) \\ \mathcal{P}_1''' = \left(0, -\dfrac{120L_{offset}}{T^3}\right) \\ \mathcal{P}_2''' = \left(0, \dfrac{60L_{offset}}{T^3}\right) \end{cases}. \tag{25}$$

The control points for each alternative collision-free reference line curve are:

$$\begin{cases} \mathcal{P}_0 = (s_0, q_0) \\ \mathcal{P}_1 = \left(\frac{v_0 T}{5} + s_0, q_0\right) \\ \mathcal{P}_2 = \left(\frac{a_s T^2}{20} + \frac{2v_0 T}{5} + s_0, q_0\right) \\ \mathcal{P}_3 = \left(\frac{3a_s T^2}{20} + \frac{3v_0 T}{5} + s_0, q_0 + L_{offset}\right) \\ \mathcal{P}_4 = \left(\frac{3a_s T^2}{10} + \frac{4v_0 T}{5} + s_0, q_0 + L_{offset}\right) \\ \mathcal{P}_5 = \left(\frac{a_s T^2}{2} + v_0 T + s_0, q_0 + L_{offset}\right). \end{cases} \tag{26}$$

By adjusting L_{offset} and a_s within the strategy set, distinct Bézier curves can be generated. Fig. 7 illustrates these alternative collision-free reference lines based on different lane-changing and braking strategies.

Let $t = k\Delta t$, where $k \in \mathbb{N}$ and $\Delta t = 0.1s$. By substituting different values into the curve formula $B(k\Delta t)_{L_{offset},a_s}$, path points for each future time interval are obtained. These points are then compared with the reachable set of the ego vehicle to compute the risk value for each curve.

Four variables are defined: cumulative risk $LaneRisk_{sum}$, minimum risk $LaneRisk_{min}$, maximum risk $LaneRisk_{max}$, and curve's value $LaneValue$. The risk value at each point in the probabilistic reachable set is denoted by $\overline{\mu}_k\left(B\left(k\Delta t\right)_{L_{offset},a_s}\right)$, and the following relationships hold:

$$LaneRisk_{sum} = \sum \overline{\mu}_k\left(B\left(k\Delta t\right)_{L_{offset},a_s}\right), \tag{27a}$$

$$LaneRisk_{min} = \min\left(\overline{\mu}_k\left(B\left(k\Delta t\right)_{L_{offset},a_s}\right)\right), \tag{27b}$$

$$LaneRisk_{max} = \max\left(\overline{\mu}_k\left(B\left(k\Delta t\right)_{L_{offset},a_s}\right)\right), \tag{27c}$$

$$LaneValue = \alpha_0 LaneRisk_{sum} + \alpha_1 L_{offset} + \alpha_2 a_s, \tag{27d}$$

where $\alpha_0, \alpha_1, \alpha_2$ are the coefficients for the curve's value. Curves with excessively high $LaneRisk_{max}$ are excluded, and from the remaining curves, the one with the lowest $LaneRisk_{sum}$ is selected as a candidate curve. If multiple curves share the same $LaneRisk_{sum}$, the one with the smallest $LaneRisk_{min}$ is chosen. If the above process fails to produce a unique curve, the optimal selection is based on the $LaneValue$ of the remaining curves. Here, the RPD offers an advantage over traditional reachable sets, as deriving risk values directly from occupancy probabilities is both simpler and sufficiently reliable.

D. Controlling Method Based on MPC

As described in [33] and [34], the nonlinear optimization process requires significant computing power. Therefore, this work adopts MPC based on Linear Time-Varying systems (LTV-MPC) for planning and control. By discretizing the yaw dynamics model using the Euler formula, the model's state-space equation can be derived as:

$$X_{k+1} = AX_k + BU_k, \tag{28}$$

where A is the state matrix, B is the input matrix, and $X_k = \left[s, v_s, q, v_q, \varphi, \dot{\varphi}\right]$ represents the state vector consisting of the vehicle's longitudinal position s, longitudinal velocity v_s, latitudinal position q, latitudinal velocity v_q, yaw angle φ, and yaw rate $\dot{\varphi}$. $U_k = [a_s, \delta]$ is the control vector, containing the longitudinal acceleration a_s and front-wheel steering angle of the vehicle δ.

To incorporate the vehicle's coordinates in the global frame, the state equation is discretized, resulting in the state-space equation:

$$X_{k+1} = \widetilde{A}X_k + \widetilde{B}U_k, \tag{29}$$

The discretized state and input matrices are $\widetilde{A}$ and $\widetilde{B}$. The values for $\widetilde{A}$ and $\widetilde{B}$ are given by the following expressions:

$$\widetilde{A} = \begin{bmatrix} 1 & \Delta t & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1+\Delta t\cdot S_{44} & \Delta t\cdot S_{45} & \Delta t\cdot S_{46} & 0 & 0 & \\ 0 & 0 & 0 & 0 & 1 & \Delta t & 0 & 0 & \\ 0 & 0 & 0 & \Delta t\cdot S_{64} & \Delta t\cdot S_{65} & 1+\Delta t\cdot S_{66} & 0 & 0 & \\ 0 & \Delta t\cdot\cos\varphi & 0 & -\Delta t\cdot\sin\varphi & 0 & 0 & 1 & 0 & \\ 0 & \Delta t\cdot\sin\varphi & 0 & \Delta t\cdot\cos\varphi & 0 & 0 & 0 & 1 & \end{bmatrix}$$

$$\widetilde{B} = \begin{bmatrix} 0 & 0 \\ \Delta t & 0 \\ 0 & 0 \\ 0 & \Delta t\cdot C_{42} \\ 0 & 0 \\ 0 & \Delta t\cdot C_{62} \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \tag{30}$$

the various S-terms and C-terms are coefficients based on vehicle dynamics parameters like tire cornering stiffness, vehicle mass, and moment of inertia.

1) Cost Function and Penalization: In MPC, an important step is selecting the appropriate state and input variables to minimize the computational burden. A balance must be struck between computational complexity and trajectory smoothness. High-dimensional matrices can lead to computational difficulties, while low-dimensional matrices can cause suboptimal vehicle trajectories.

In the cost function, the selected state and input variables (e.g., longitudinal and latitudinal coordinates, velocities, and

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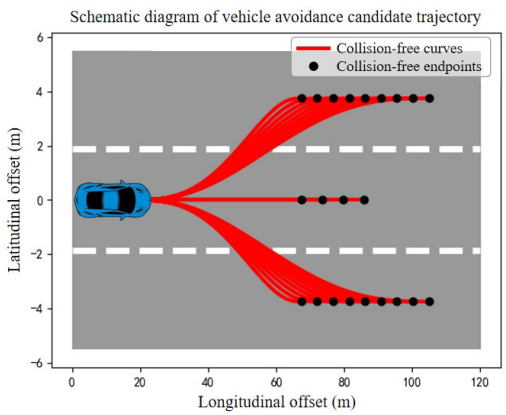

Fig. 7. Diagram of alternative collision-free reference lines.

steering angles) are penalized to ensure optimal control. The state vector Z_k is defined as:

$$Z_k = C_z X_k + D_{zu} U_k, \tag{31}$$

The difference between successive control inputs is also penalized:

$$\Delta U_k = \Delta U_k - \Delta U_{k-1}, \tag{32}$$

where C_z and D_{zu} are the matrix of the states and the input distributions. Constraints on the system's behavior are incorporated as follows:

$$f_k = C_f \cdot x_k + D_{fu} \cdot u_k, \tag{33}$$

$$\underline{f} \leq f_k \leq \overline{f}, \tag{34}$$

where $f_k = \left[v_x, v_q, \dot{\varphi}, a_x, \delta\right]$ is the vector containing the vehicle's longitudinal velocity, latitudinal velocity, angular velocity, longitudinal acceleration, and steering angle. C_f and D_{fu} is the matrix of the distribution of the state and the input. $\overline{f}$ and $\underline{f}$ define the bounds on these states.

2) MPC System and Time Complexity: The entire system and the cost function are formulated by stacking the relevant matrices over the prediction horizon, resulting in:

$$X = \overline{A} X_0 + \overline{B} U, \tag{35}$$

$$f = \overline{C_f} X_0 + \overline{D_{fa}} U, \tag{36}$$

$$Z = \overline{C_Z} X_0 + \overline{D_{Zu}} U. \tag{37}$$

The cost function, J, is then expressed as:

$$J = \left(\overline{C_Z} X_0 + \overline{D_{Zu}} \overline{U} - \overline{Z}\right)^T Q \left(\overline{C_Z} X_0 + \overline{D_{Zu}} \overline{U} - \overline{Z}\right) + \overline{\Delta U}^T P U \overline{\Delta U}. \tag{38}$$

The objective is to minimize the difference between the controlled trajectory and the planned path where $\overline{Z}$ represents the state information of the selected path at each time interval. The cost function could be simplified further as:

$$J = \overline{U}^T \overline{D_{Zu}}^T Q \overline{D_{Zu}} \overline{U} + 2\left(X_0^T \overline{C_Z}^T Q \overline{D_{Zu}} - \overline{Z}^T Q \overline{D_{Zu}} Q\right) \overline{U} + \overline{\Delta U}^T P U \overline{\Delta U} + X_0^T \overline{C_Z}^T Q \overline{C_Z} X_0 - 2\overline{Z}^T Q \overline{C_Z} X_0 + \overline{Z}^T Q \overline{Z}. \tag{39}$$

Finally, after simplifying the cost function, the time complexity of the MPC algorithm is:

$$O\left(N_p \times n_{ob} + \left(N_p \times n_u\right)^3\right), \tag{40}$$

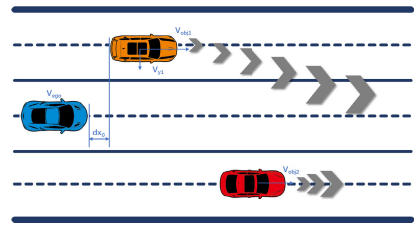

Fig. 8. The diagram of the cut-in scenario.

where N_p is the number of predictive time steps, n_{ob} is the number of obstacles, and n_u is the dimension of the control vector. While the number of time steps and control dimensions are fixed, the algorithm's complexity can become more sensitive to the number of obstacles, which can introduce significant computational overhead.

IV. Experimental Study

Inspired by [33] and [34], the proposed method was experimentally tested under simulation and vehicle conditions. The comparative experiments focused on cut-in and cut-out scenarios on straight roads, which are recognized as representative test cases for AV's SOTIF in [35]. By comparing with the baseline EM Planner method, the framework's superiority and adaptability in emergency collision avoidance were verified. In the vehicle experiment, the proposed method was deployed on a real vehicle for testing, verifying its feasibility under real road conditions.

A. Simulation Experiments

The simulation experiment will test our risk avoidance efficiency, performance, and adaptability. All simulation experiments will be conducted in a Python environment, and the test scenarios will be collected from our own dataset.

1) Comparison Study: In the comparison experiment, the proposed method was evaluated against the baseline method in cut-in and cut-out scenarios. In the cut-in scenario (Fig. 8), an obstacle vehicle ahead and to the left of the ego vehicle attempts to cut into its path, while another obstacle vehicle on the right blocks the option to change lanes. This situation notably elevates the collision risk, posing significant challenges to emergency maneuvering strategies.

Specifically, the initial speed of the ego vehicle was $V_{ego} = 15\ m/s$, with obstacle vehicle 1 (located in the left front of the ego vehicle) cutting into the path at $V_{obs_1} = 12\ m/s$, and obstacle vehicle 2 drove along the center of the right lane at a constant velocity $V_{obs_2} = 15\ m/s$. Fig. 9-(a) and 9-(b) show the trajectories and directions of the ego vehicle and obstacle vehicles under two methods, while Fig. 10-(a) to 10-(d) compare the dynamic performance in terms of lateral stability and risk elimination efficiency.

In the cut-in scenario, the ego vehicle switched from the middle to the left lane under both methods to avoid the collision with obstacle vehicle 1. However, the two methods show different results. As seen in the comparison between Fig. 9-(a) and 9-(b), the proposed method allowed the ego

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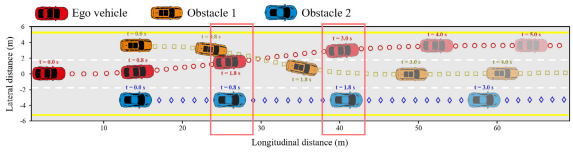
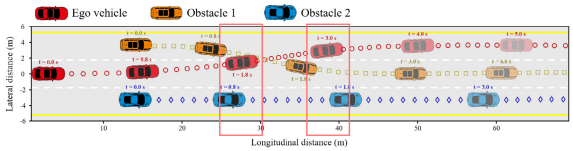
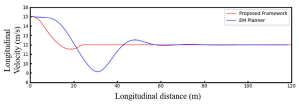


(a) Emergency operation of the proposed method

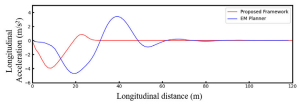


(b) Emergency operation of the baseline method

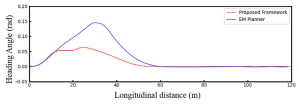
Fig. 9. Cut-in scenario simulation: Trajectory comparison between the proposed and baseline methods.



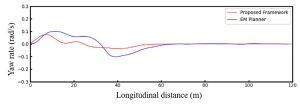
(a) Longitudinal Velocity



(b) Longitudinal Acceleration



(c) Heading Angle



(d) Yaw Rate

Fig. 10. Cut-in scenario simulation: Vehicle performance comparison between the proposed and baseline methods.

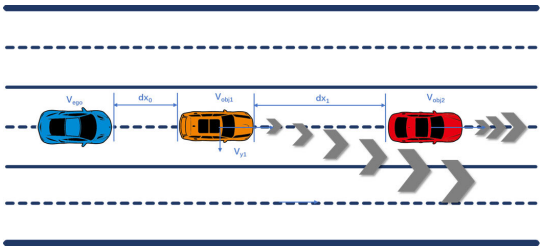
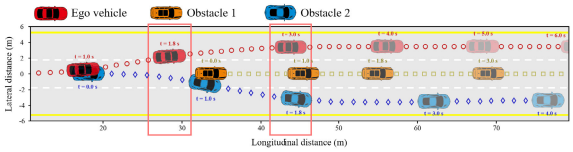


Fig. 11. The diagram of the cut-out scenario.

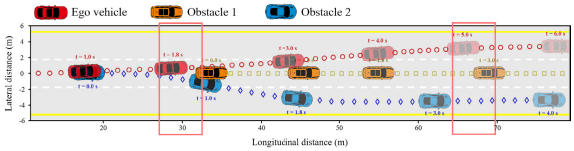
vehicle to evade earlier and recover to normal driving faster, demonstrating better performance in this scenario.

Additionally, as shown in Fig. 10-(a) to 10-(d), the proposed method exhibited smoother velocity and yaw angle rate variations. After $T = 4\ s$, the EM Planner method oscillations appeared as the vehicle's heading was still being adjusted. In contrast, the state of the ego vehicle under the proposed method converged quickly, further highlighting the advantages of the proposed method.

In the cut-out scenario, as shown in Fig. 11, two obstacle vehicles existed before the ego vehicle. The vehicle in the middle temporarily blocked the view of the vehicle in front, which the ego vehicle could not detect. The middle vehicle cut to the right at some point, exposing the leading vehicle ahead. In this case, the collision risk increased due to the leading vehicle's significantly lower speed than the ego vehicle and

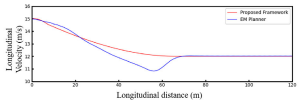


(a) Emergency operation of the proposed method

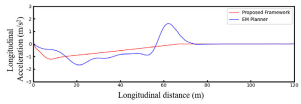


(b) Emergency operation of the baseline method

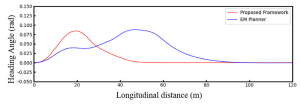
Fig. 12. Cut-out scenario simulation: Trajectory comparison between the proposed and baseline methods.



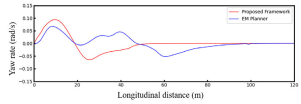
(a) Longitudinal Velocity



(b) Longitudinal Acceleration



(c) Heading Angle



(d) Yaw Rate

Fig. 13. Cut-out scenario simulation: Vehicle performance comparison between the proposed and baseline methods.

the sudden appearance of the obstacle in the field of view, posing a similar challenge to emergency avoidance strategies and control algorithms.

The initial conditions for the scenario were as follows: the ego vehicle traveled at $V_{ego} = 15\ m/s$, with obstacle vehicle 1 positioned 15 m ahead and moving at the same speed $V_{obs_1} = 15\ m/s$. Obstacle vehicle 2 was located 30 m ahead of the ego vehicle and traveled at a slower speed of $V_{obs_2} = 12\ m/s$. The trajectories of the ego and obstacle vehicles under the two methods are shown in Fig. 12-(a) to 12-(b). Additionally, Fig. 13-(a) to 13-(d) compares the methods' dynamic performance in terms of lateral stability and risk elimination efficiency.

In the cut-out scenario, the ego vehicle again adopts a strategy of switching from the middle to the left lane, which is reasonable due to the obstruction of the obstacle vehicles. Comparing Fig. 12-(a) and 12-(b), the proposed method responds to the risk earlier and recovers to normal driving faster. As shown in Fig. 13-(a) to 13-(d), the proposed method also demonstrates smoother velocity and yaw angle rate changes in this scenario.

2) Adaptability Study: In Chapter III, the generalization of the proposed method was qualitatively discussed but lacked quantitative validation under varying conditions. To address this, a quantitative generalization test based on [35] was conducted in the cut-in scenario depicted in Fig. 8. By fixing

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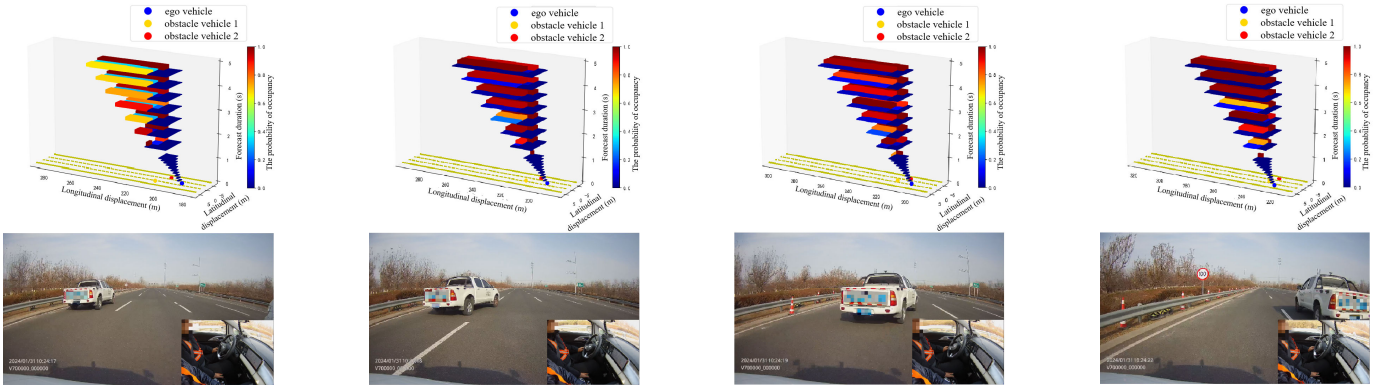
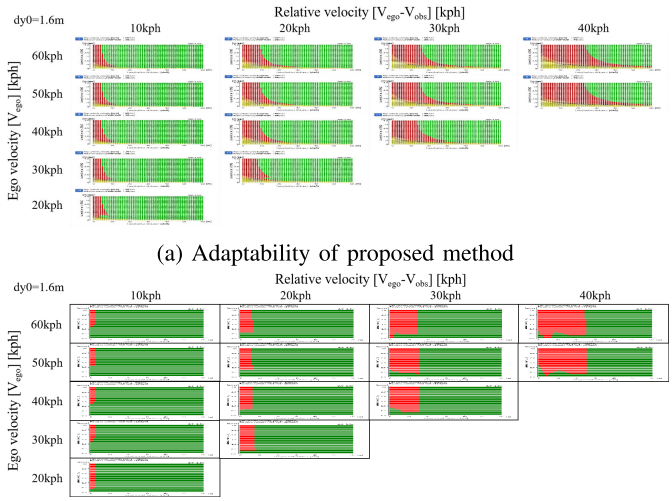


(a) T = 17.65s (b) T = 18.55s (c) T = 19.45s (d) T = 22.65s

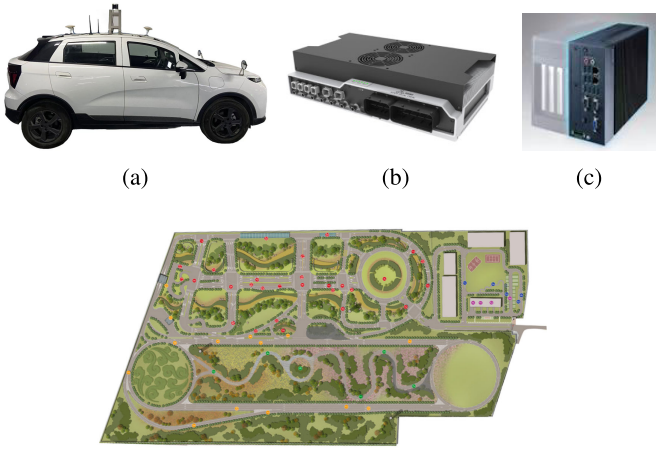
Fig. 14. The AV's RPD at each phase in the cut-in scenario.



(a) Adaptability of proposed method

(b) Adaptability of ALKS

Fig. 15. Comparison of adaptability in cut-in scenarios.



(a) (b) (c) (d)

Fig. 16. Autonomous vehicle platform and Testing Base: (a) Test vehicle, (b) Computer for planning, (c) Computer for controlling, (d) ICV testing base.

the lateral distance $dy0 = 1.6\ m$ between the ego vehicle and obstacle vehicle 1, the test varied the ego vehicle speed $Ve0$, obstacle vehicle speed $Vo0$, longitudinal relative distance $dx0$, and obstacle vehicle 1's lateral speed Vy during the cut-in.

B. Vehicle Experiments

The results were compared with the Advanced Lane Keeping System (ALKS) from [35], as shown in Fig. 15. Fig. 15-(a) presents the ALKS results, while Fig. 15-(b) shows the performance of the proposed method. Under all tested conditions, the proposed method demonstrated superior adaptability, highlighting its effectiveness in the cut-in scenario. In this experiment, all test vehicles were equipped with On-Board Units (OBUs) while the roads were covered with Road-Side Units (RSUs), enabling reliable V2X communication between the vehicles, infrastructure, and other traffic participants.

In the context of vehicle experiments, The proposed method has been deployed on autonomous driving vehicles and tested at the National ICV Innovation Park in Yizhuang, China, as depicted in Fig. 16-(d). Inspired by [36], this work employs a two-stage computation approach separately using the two computing platforms shown in Fig. 16-(b) and Fig. 16-(c) to improve efficiency, achieving a comparable frequency of 10 Hz. Unlike [36], which focuses on multi-vehicle coordination, our method emphasizes single-vehicle optimization with RPD-based probabilistic risk assessment, offering superior adaptability in mixed-traffic environments.

In cut-in scenario, the ego vehicle traveled at a speed of $V_{ego} = 30\ km/h$. Obstacle vehicle 1 attempted to cut into the lane of AV, traveling at a speed of $V_{obs_1} = 20\ km/h$. The obstacle vehicle 2 was traveling at $V_{obs_2} = 30\ km/h$. When $TTC \leq 5.5s$, proposed method was activated and directed the ego vehicle to the left.

The ego vehicle's RPD at different stages is shown in Fig. 14. At $T = 17.65\ s$, obstacle vehicle 1 began cutting into the middle lane, completing the cut-in at $T = 18.55\ s$. During this period, the probability of the left lane being occupied decreased, while the middle lane's occupancy likelihood increased. In response, as shown in Fig. 14-(c) to 14-(d), the ego vehicle initiated a left lane change to avoid the cutting-in obstacle. By effectively evaluating surrounding risks with

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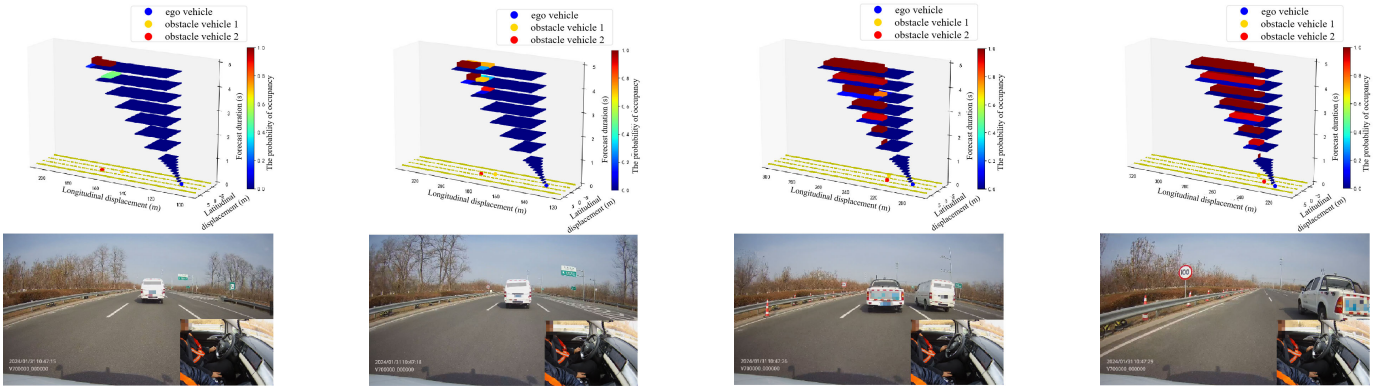
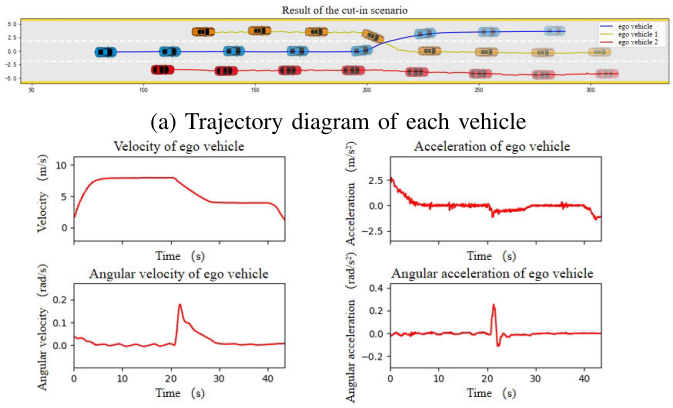


(a) T = 10.8s (b) T = 13.8s (c) T = 21.2s (d) T = 24.4s

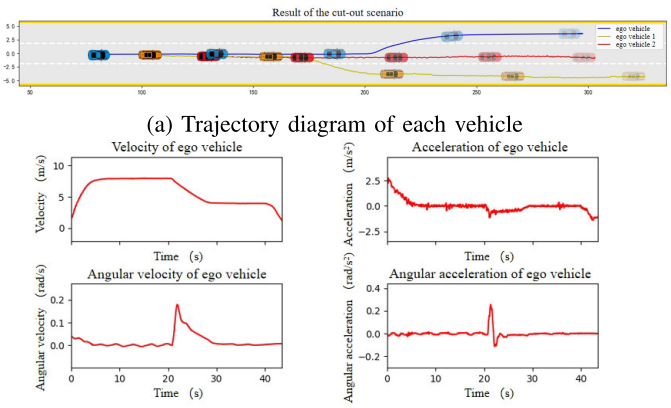
Fig. 17. The AV's RPD at each phase in the cut-out scenario.



(a) Trajectory diagram of each vehicle

(b) The AV's velocity, acceleration, angular velocity, and angular acceleration in the cut-in scenario

Fig. 18. Vehicle test in cut-in scenario.



(a) Trajectory diagram of each vehicle

(b) The AV's velocity, acceleration, angular velocity, and angular acceleration in the cut-out scenario

Fig. 19. Vehicle test in cut-out scenario.

RPD and leveraging MPC for decision-making, the proposed method ensured both safety and stability.

Fig. 18-(a) represents each vehicle's trajectory diagram during the test. Fig. 18-(b) illustrates the vehicle states over time during the experimental process. These curves show variations in the ego vehicle's speed, acceleration, and angular velocity.

In cut-out scenario, the ego vehicle traveled at $V_{ego} = 30\ km/h$, matching the speed of obstacle vehicle 1, while obstacle vehicle 2 moved slower at $V_{obs_2} = 20\ km/h$. As obstacle vehicle 1 approached the slower obstacle vehicle 2, it maneuvered right to avoid a collision. When the TTC between the ego vehicle and obstacle vehicle 2 dropped below 3.5 seconds, the emergency evasion system intervened, directing the ego vehicle to the left lane.

In this scenario, Fig. 17 shows the RPD distribution of the ego vehicle across two phases. From Fig. 17-(a) to 17-(b), obstacle vehicles maneuver: at $T = 10.8\ s$, obstacle vehicle 1 approached obstacle vehicle 2 and moved into the right lane by $T = 13.8\ s$ to avoid a collision, shifting the ego vehicle's RPD risk from the middle to the right lane. From Fig. 17-(c) to 17-(d), after emergency evasion activated, the ego vehicle maneuvered left, starting at $T = 15.1\ s$, initiating the lane change at $T = 21.2s$, and completing it at $T = 24.4\ s$. During this time, the RPD remained relatively stable.

The trajectory of each vehicle is shown in the Fig. 19-(a). For the abovementioned experiments, we also analyzed the temporal evolution of vehicle states, as depicted in Fig. 19-(b). These curves illustrate the variations in the ego vehicle's speed, acceleration, angular velocity, and angular acceleration.

The current study focuses on high-speed roadway scenarios. In the future, special tests will be carried out for low-speed high-curvature environments to verify the effectiveness of the method under complex steering conditions.

V. Conclusion And Prospect

This paper introduces an emergency avoidance framework comprising a collision-free path planning method based on the RPD and an avoidance control approach utilizing MPC. The framework begins by calculating the reachable sets of the AV and surrounding obstacles, leveraging the IMM to predict the probabilities of obstacles occupying different lanes over time. Then, an RPD map is constructed to represent the probability distribution of the AV's reachable sets relative to obstacles. Collision-free reference trajectories are then generated within the RPD using a Bézier curve-based strategy. Finally, an

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MPC-based control method directs the AV to a safe area. This framework has been successfully deployed and tested in various scenarios, demonstrating its ability to ensure AV safety while maintaining stability in real world conditions. Meanwhile, this framework has further improved the safety of intelligent transportation systems during emergencies by accurately predicting dynamic obstacles and optimizing path planning, providing a novel and practical solution to ensure their safety.

Future work could refine the IMM algorithm to improve position probability precision in complex scenarios. Furthermore, the computational efficiency of the framework is sensitive to the number of obstacles (Eq. (40)). Inspired by the methodology in [36], we will focus on enhancing adaptability to dense traffic by addressing computational bottlenecks. The proposed method currently focuses primarily on preventing catastrophic hazards, with limited consideration for adherence to traffic rules or common-sense driving behaviors. Future work will address this limitation by optimizing the path-planning algorithm to ensure trajectories comply with traffic regulations and natural driving conventions. Notably, the current kinematic model overlooks tire dynamics and lateral slip under extreme steering inputs, limiting its applicability to low-speed, high-curvature scenarios. To address this, integrating dynamic models and validating the framework in curved urban roads will be explored.

Meanwhile, to improve generalization and scene coverage, particularly for unstructured roads and complex constraints such as traffic rules, integrations with machine learning will be explored. For instance, reinforcement learning could optimize discrete strategy selection, while data-driven models may enhance environmental comprehension for probabilistic occupancy estimation. These advancements will strengthen the theoretical and practical foundations of autonomous driving technologies, driving progress toward more intelligent and safer transportation systems.

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